

Omkar Sudhir Patil

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Education & Current Position

University of Florida

Postdoctoral Research Associate

2023–Present

University of Florida

PhD in Mechanical Engineering

2019–2023

Dissertation: Implicit and Deep Learning-Based Control Methods for Uncertain Nonlinear Systems

Advisor: Dr. Warren Dixon

Indian Institute of Technology Delhi

B.Tech in Production and Industrial Engineering

2014–2018

Selected Publications

Book: O. Patil, E. Griffis, and W. E. Dixon, *Deep Learning-Based Adaptive Control: A Lyapunov-Based Approach*. In development with Springer, 2025.

IEEE TAC 2025: O. Patil, R. Kamalapurkar, and W. Dixon, "A saturated RISE controller with exponential stability guarantees," *IEEE Trans. Autom. Control*, 2025.

IEEE Access 2025: O. Patil, D. Le, E. Griffis, and W. Dixon, "Lyapunov-based deep residual neural network (ResNet) adaptive control," *IEEE Access*, 2025.

IEEE TAC 2022: O. Patil, R. Sun, S. Bhasin, and W. E. Dixon, "Adaptive control of time-varying parameter systems with asymptotic tracking," *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4809–4815, 2022.

IEEE CSL 2022: O. Patil, D. Le, M. Greene, and W. E. Dixon, "Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network," *IEEE Control Syst Lett.*, vol. 6, pp. 1855–1860, 2022.

Automatica 2025: W. Makumi, O. Patil, and W. Dixon, "Lyapunov-based adaptive deep learning for approximate dynamic programming," *Automatica*, vol. 180, p. 112462, 2025.

Recognition & Service

2023: Graduate Student Research Award, University of Florida Department of Mechanical and Aerospace Engineering

2018: BOSS Award for Best Hardcore Experimental Project, IIT Delhi

Reviewing: IEEE Transactions on Automatic Control, Automatica, IEEE Control Systems Letters, NeurIPS, ICLR

Conferences: IEEE Conference on Decision and Control, American Control Conference, IFAC World Congress

Research Impact

Publications: 25+ journal papers (published/under review) in top-tier venues including IEEE TAC, Automatica

Conferences: 12+ conference presentations at premier control systems conferences

Collaboration: Extensive multi-author research demonstrating strong collaborative abilities